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Magnetic Reversal Sequence in the Upper Portion of the Chinle Formation, Montoya, New Mexico

ABSTRACT

An investigation of the magnetic reversal stratigraphy in the Redonda Member and the underlying shale member of the Chinle Formation at Montoya, New Mexico, reveals the presence of at least eight separate polarity intervals in the upper 133 m of the Chinle Formation. At Lucianna Mesa south of Montoya the contact between the Redonda Member and the underlying unnamed shale member occurs 3 m above a transition from normal (below) to reversed polarity (above), the reversal sequence within the Redonda having eight polarity intervals. Measurement of samples collected from the same stratigraphic interval on Redonda Mesa 30 km to the east serves as a check on a portion of this reversal sequence. The Chinle Formation in eastern New Mexico is considered to have been contemporaneous with the Newark group in the eastern United States on the basis of vertebrate paleontology. Published investigations of the Newark group have revealed only normal magnetic polarities. Because the magnetic field reversed for the entire earth the Newark cannot exhibit different polarities from those in the Chinle if the two units acquired their magnetization at the same time. Thus, there are inconsistencies between the paleontologic and paleomagnetic data for the two rock units. These inconsistencies are best answered by assuming the Chinle in New Mexico is partially or completely older than the Newark. The latter explanation is improbable when carried to the extent of assigning the Chinle a Middle Triassic age as proposed by McElhinny and Burek (1971). Until the temporal relation between the Chinle and Newark is clarified, the Newark group should not be used to define a reversal chronology.

A mean paleomagnetic pole position at 79° E. long, 58° N. lat was calculated for approximately the upper half of the Chinle Formation at Montoya. Pole positions for two well-demagnetized polarity intervals (one from each member studied) have been combined with other published Upper Triassic pole positions (demagnetized) to obtain a mean Upper Triassic paleomagnetic pole position at 89° E. long, 62.5° N. lat. The similarity of this mean and of the individual Chinle poles to the published Lower Triassic pole determinations based on demagnetized samples indicates little, if any, latitudinal polar wander–continental drift relative to North America during most if not all of the Triassic.

INTRODUCTION

Recent papers by Khramov (1958), Picard (1964), Burek (1967, 1970), Helsley (1969), and van Montfrans (1971) point out that magnetic polarity reversal sequences may be used advantageously in the solution of both regional and global stratigraphic problems. Reversal sequences determined from volcanic or sedimentary units are also important in that they provide a check on reversal sequences derived from sea floor magnetic anomaly patterns (Evans, 1971; Watkins and others, 1971; Watkins and Goodell, 1967). For example, Foster and Opdyke (1970) determined polarity sequences from cores recovered in the Pacific that were in excellent agreement with the polarity sequence determined from sea floor anomalies (Vine, 1968; Pitman and Heirtzler, 1966).

Determination of the polarity reversal sequence in the Redonda Member of the Chinle Formation of eastern New Mexico was undertaken during the summers of 1970 and 1971 with the intention of later developing a

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"standard" polarity reversal sequence for the Upper Triassic of this area that could be used to determine correlations between the Chinle and the Dockum Group in Texas and the Upper Triassic of the eastern United States. During the second field season, a number of samples were collected in an unnamed shale member of the Chinle that underlies the Redonda Member and a second section was collected at Redonda Mesa, the type locality of the Redonda Member of the Chinle Formation. This second section served as a check on the reversal sequence derived from the first stratigraphic section sampled. This second section is discussed in the Appendix.

GEOLOGY AND PREVIOUS WORK

The Redonda Member of the Chinle was first named and described by Dobrovolny and others (1946) and later described in more detail by Griggs and Read (1959) who raised it to formational status and tentatively assigned it to a Late Triassic (Glen Canyon) age. More recent mapping by Dane and Bachman (1965) places the Redonda back in the Chinle Formation.

The Redonda consists of alternating 1 to 10 ft thick beds of fine-grained sandstones, siltstones, and mudstones. Ripple marks, small scale crossbeds, and laminar bedding are common, although many beds show no sedimentary structures other than the results of intense burrowing. Mudcracks and loadcasts also occur within the Redonda. Dobrovolny and others (1946) associated calcareous and fine-grained clastics within the Redonda Member with a fluvial-deltaic depositional environment of the unnamed shale member and suggested a local lacustrine origin for the Redonda. The unit has a distinct brick red color that sets it off from the darker red of the underlying unnamed shale member of the Chinle Formation and the yellow to white sandstone of the Chinle Formation and the yellow to white sandstone of the unconformably overlying Entrada Formation. The unit is exposed on the lower slopes of numerous mesas in the region around Tucumcari, New Mexico.

Previous paleomagnetic work was done by Graham (1955) who collected 17 samples at Lucianna Mesa, the original locality studied in this paper. Graham did not demagnetize his

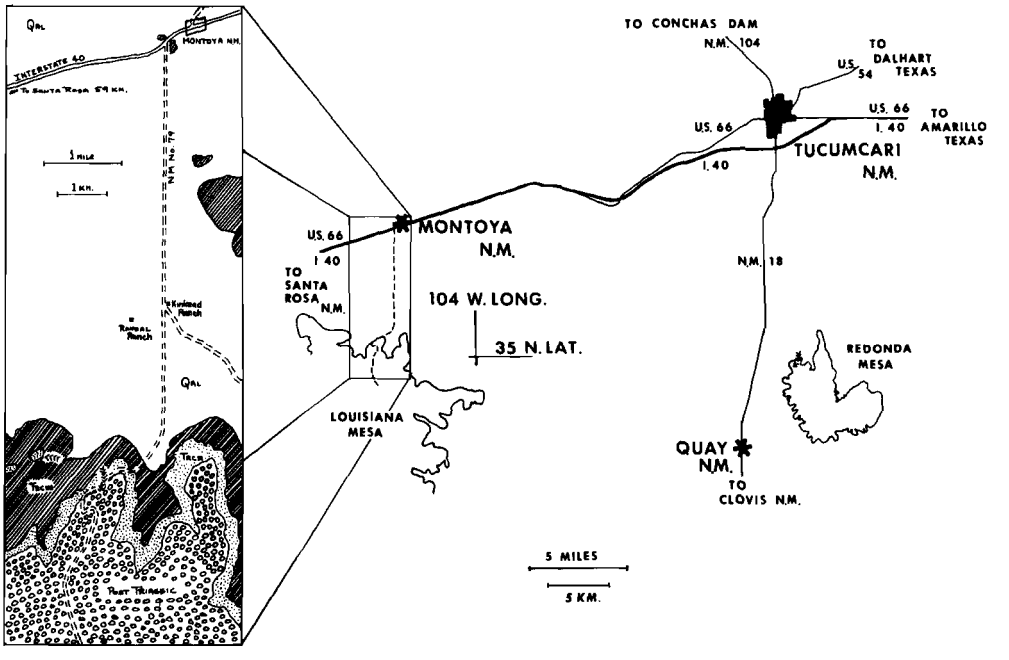


Figure 1. Geographic location of the sampling sites. Enlargement is geologic map showing the outcrop of unnamed shale member and Redonda Member of the Chinle Formation at Lucianna Mesa. /// = unnamed shale member; ∵∴∵ = Redonda member; ⁘ = post-Triassic rocks; No symbol = Quaternary alluvium; >>>> = sampled sections (both localities).

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samples or attempt to define a reversal sequence. Data from Graham's Figure 7 (1955, p. 348) were used by Cox and Doell (1960) to obtain a pole position for the Redonda at 77° N. lat, 23° W. long. This pole agrees well with that derived from the average natural remanent magnetization (NRM) of 240 samples from this study (77° N. lat, 27° W. long).

SAMPLING AND MEASUREMENT

The main sampling locality is located at the base of Lucianna Mesa 8.9 km south of Montoya, New Mexico, on state highway 79 (lat 35°02′ N., long 104°05′ W.). To check a portion of the magnetic reversal sequence derived from this locality, a second stratigraphic section was collected from the north side of Redonda Mesa 19 km south of Tucumcari, New Mexico (lat. 34°59′ N., long 103°44′ W.). Sample localities are shown in Figure 1. Samples were collected with a hand-held gasoline drill and were oriented with a magnetic compass. The samples were collected in continuous stratigraphic sequence, the average interval between samples being 48 cm for the Lucianna Mesa section and 89 cm for the section at Redonda Mesa.

Upon return to the laboratory, all samples were cut to 2.5 cm cylinders and measured on a PAR SM-1 spinner magnetometer.

NRM AND THERMAL DEMAGNETIZATION

General Procedures and Discussion of Sample Behavior

The natural remanent magnetization (NRM) of the samples from Lucianna Mesa is shown in Figures 2 and 3. The average inclination of the samples is 56.5° and the average declination is 15°. The field at the sampling locality now has an inclination of 62° and a declination of approximately 12°. The average NRM inclination and declination of the samples reflects a large secondary component of magnetization in the direction of the earth's present magnetic field.

In addition to this component, there is a

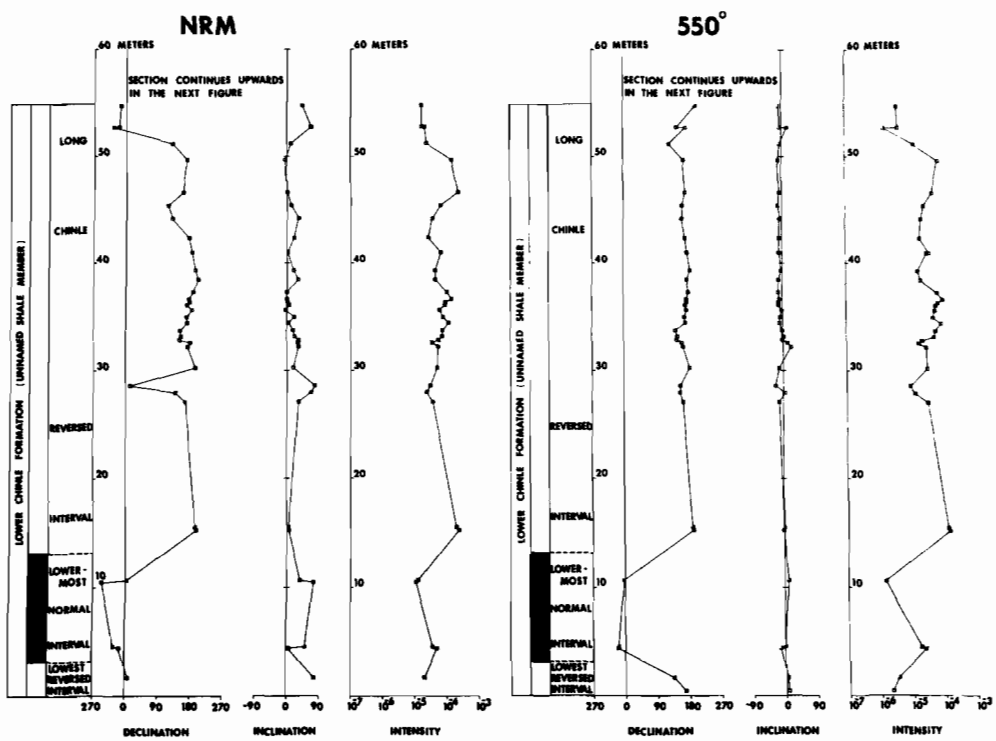


Figure 2. Stratigraphic plots of NRM and 550° C demagnetized sample declination, inclination, and intensity for lower portion of Lucianna Mesa section. Base of section (0 m) coincides with base of the lowest outcrop.

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very low stability component acquired during routine handling and preparation of the samples in the laboratory that is probably due to a viscous moment. This low stability component caused some samples to drift noticeably during measurement on the PAR SM-1. Storing the samples within a magnetically shielded room (<150 gammas) for a period of several days to a few weeks before measuring their NRM noticeably reduced the instability of many samples, the viscous moment having decreased in the absence of an orienting field. This strongly suggests that some of the secondary component in the direction of the earth's present field may be viscous in origin.

Thermal demagnetization was used to remove as much of the secondary component as possible. An initial group of 24 samples was

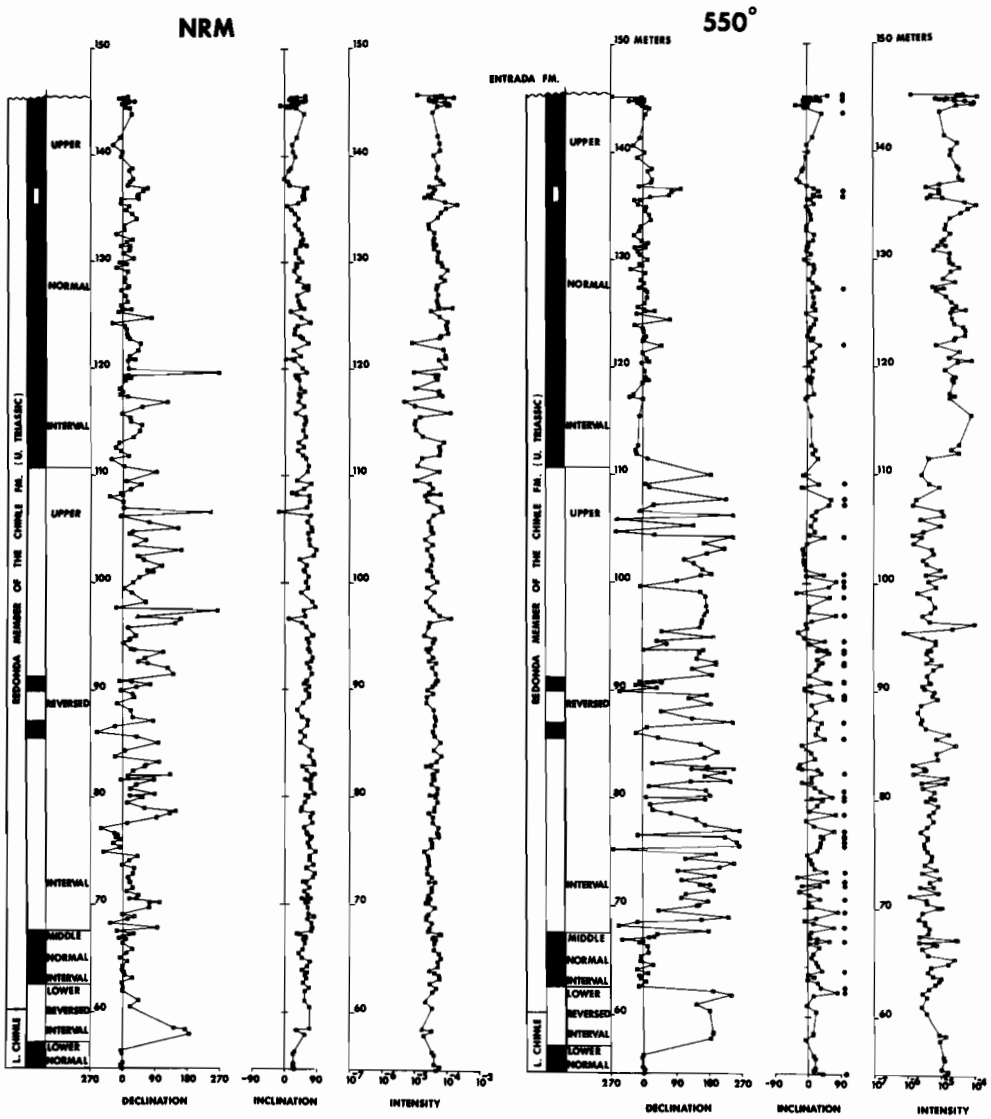


Figure 3. Stratigraphic plots of NRM and 550° C demagnetized sample declination, inclination, and intensity for upper portion of Lucianna Mesa section. Base of section coincides with base of lower normal interval. Samples marked with one dot were excluded from statistical analysis and determination of pole positions. This section is a continuation of Figure 2.

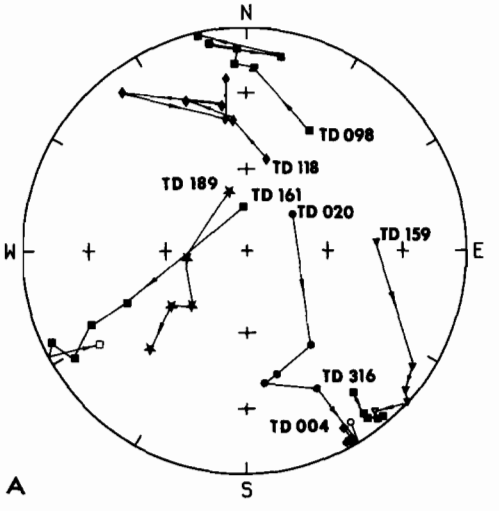
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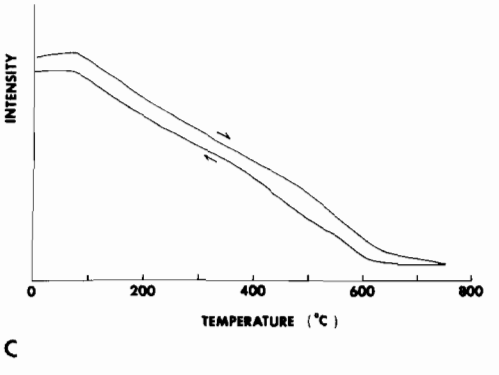
stepwise demagnetized at temperatures of 250°, 300°, 400°, 550°, and 630° C to determine the nature of the remanent magnetization. As a result of these initial stability tests, all samples were heated to 400° C and 550° C (± 30° C). Exceptions to this blanket treatment were the 12 green to white porous sandstones collected from the interval 108 to 117 m in Figure 3 which were not demagnetized. In addition, about one-third of the samples were again demagnetized at 630° C (± 10°).

The samples were demagnetized 12 at a time in a four-level noninductively wound electrical furnace. Platinum thermocouples referred to an ice bath and read on a digital voltmeter were used to record the temperature on each level of the furnace. The average length of time the samples were held at the maximum temperature was approximately 15 min. The furnace was positioned inside a pair of Helmholtz coils and was located in a magnetically shielded room. Residual field in the area of the cooling samples was ± 10 gammas or less. All samples were measured within 24 hours of heating and the maximum field exposure of any sample between heating and measurement was less than 150 gammas. Nevertheless, the two westward deviations in the direction of sample TD 118 (Fig. 4A) associated with gains in remanent intensity (Fig. 4B) indicates some samples were influenced by the small residual field. Usually this occurred at temperatures above 550° C (for example, the eastward deviation of TD 098 at 630° C).

In conjunction with demagnetization, polished sections were studied from ten of the samples. Curie balance studies were carried out on these and on two other samples. These samples represent all major lithologic variations at the sampling locality and all the polarity intervals encountered.

Magnetic minerals in the Redonda Member and the underlying unnamed shale member of the Chinle Formation occur mainly in two





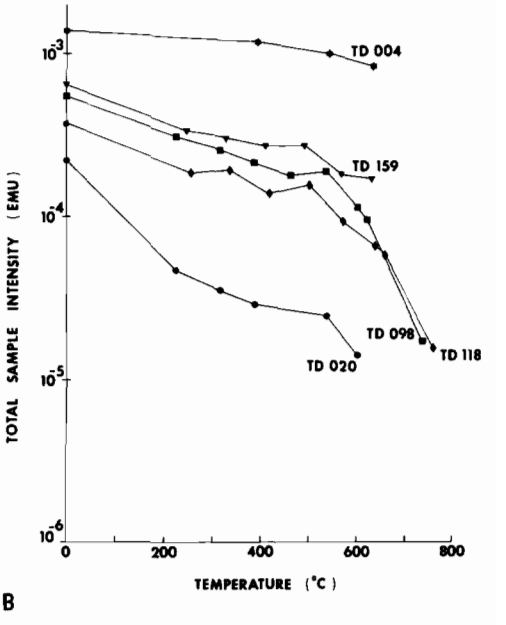


Figure 4. A. Schmidt net (equal area) plot of successive declination and inclination changes with temperature for selected samples. Solid and open symbols represent lower and upper hemispheres, respectively. B. Variation of remanent total sample intensity with temperature. C. Typical Curie balance curve for samples from the Redonda Member of the Chinle Formation. Behavior from 0° to 75° C is a property of the particular measuring instrument used and is not a property of the rock itself.

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forms, red pigment and black mineral oxides. The red pigment (hematite and some iron hydroxides) occurs irregularly as small particulate crystals and as a thin film. The red hematite pigment occasionally occurs between grain contacts as very thin (.5 mm or less), discontinuous subhorizontal seams which are devoid of coarser grains. Some heavy minerals have well-developed granular hematite coatings that appear to have grown in situ. Other occasional occurrences of pigment between grain contacts may represent growth of the pigment at the expense of the host grain, as in the case of the heavy minerals, or reddening during a late compactional stage of diagenesis.

Polished section studies indicate that the black mineral oxide grains are composed chiefly of hematite and hematite with exsolved ilmenite, although considerable ilmenite with exsolved hematite is also present. Trace amounts of magnetite and exsolved rutile are found within some of the hematite and ilmenite grains. In addition, one or two grains of martitized magnetite occur in several slides. The size of the black mineral oxide grains ranges from 25 to 55 microns, averaging about 35. The lack of inflections in the Curie balance curves at the magnetite Curie point are interpreted as indicating a low percentage of magnetite in the sediment. A typical Curie curve exhibiting an indistinct hematite Curie point is shown in Figure 4C.

With the exception of the white porous sandstones and a dark red sandstone bed in the unnamed shale member which exhibits a more rounded curve and distinct Curie point at 685° C, all the lithologies in the upper part of the Chinle Formation exhibited similarly shaped Curie curves regardless of intensity or polarity.

Comparison of plots of change in declination and inclination with temperature (Fig. 4A) and change in remanent intensity with temperature (Fig. 4B) indicates that the stable remanence (Triassic direction) is held only by those domains with very high blocking temperatures. Thermally discrete samples, such as TD 004, which lost 36 percent of its remanent intensity from 0° to 638° C, usually changed NRM direction much less than the more thermally distributed samples like TD 020, which lost 92 percent of its remanent intensity from 0° to 603° C. Parallelism of NRM direction with the stable Triassic direction is directly correlated with high thermal stability. Remanent intensity in all samples carried to temperatures of 650° C and above dropped by at least an order of magnitude. Similar behavior and combination of thermally discrete and thermally distributed components of magnetization in the same sample have been described by Jones and McElhinny (1967), who associated the origin of the thermally distributed component with severe folding, and by Irving and Opdyke (1965), who proposed either a viscous or chemical origin associated with moderate heating during burial. The Chinle Formation in eastern New Mexico has not undergone any severe folding so this cannot serve as the heat source, nor has it been deeply buried. Both viscous and chemical origins are possible. The large present-day viscous component, along with a fairly shallow burial history, led to the conclusion that the distributed component is caused by weathering effects and is chemical in origin with current weathering actually being responsible for much of the viscous component. The major support for this hypothesis is the large number of samples (47) that retain high inclinations ($>30°$) associated with the present field at temperatures of 550° C or higher (samples marked with dot in Fig. 3). The concentration of these samples within the upper reversed, middle normal, and lower reversed intervals supports this interpretation if one assumes that their lower intensities are associated with a lithology more susceptible to weathering.

The series of plots in Figure 5 show group movements of each of the normal and reversed intervals. The normal intervals are characterized by NRM groupings at shallower inclinations than the direction of the present field and move north-northeast to shallower inclinations upon demagnetization. The reversed groups are characterized by their general scatter upon demagnetization, general movement to the south and upper hemispheres, and noticeable lack of points at shallow inclinations toward the north-northeast. In both cases scatter at the 630° C demagnetization step increased slightly as some remanent intensities decreased enough to be overprinted by the small TRM acquired during cooling in a 10 gamma field. Group declination and inclination plots at 630° C are further biased by the failure to demagnetize all the samples at this temperature, so plots for this temperature are not included. It was evident that a high percentage of the reversed samples of the upper

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reversed interval were still moving to the south at 630° C and that some intervals, particularly the upper and lower reversed intevals in the Redonda Member, are still incompletely demagnetized at the 550° C demagnetization step. Twelve samples collected at Lucianna Mesa from the long Chinle reversed interval late in the course of this study were not demagnetized at 400° C. This causes an apparent slight increase in spread between the 400° C and 550° C plots for this interval.

Figures 2 and 3 show stratigraphic plots of the declination and inclination of all samples demagnetized at 550° C. A definite reversal sequence is clearly present within the Redonda Member and the upper portion of the unnamed shale member of the Chinle Formation, although some intervals are quite scattered. Within the uppermost normally magnetized interval is a 1.3-m-thick, anomalous interval. In the NRM section this interval is represented by three samples with distinct deviations in declination, inclination, and intensity. At 550° C the declination increased from around 60° to 90°; inclinations are slightly high and intensities are lower than those of the upper normal interval, and are similar to the intensities of the other reversed intervals in the Redonda. This zone may be due to incomplete demagnetization of the samples or may be due to lithologic variations, or a period of anomalous magnetic field. It has not been included in the pole determinations.

Intensities of the demagnetized samples of the upper normal itself are higher than those of the upper reversed, middle normal and lower reversed intervals (Table 1). Other authors have noted the increase in the ratio of intensity of normal to reversed samples during demagnetization. Helsley (1969) suggested this implied that the origin of the intensity of the reversed samples was different from the origin of the normal samples, and was probably due to lithologic differences. One possible explanation for the low intensities in the Redonda is that the samples have acquired a small, highly stable secondary chemical remanence in the direction of the present-day field or possibly in the direction of the Jurassic field during pre-Entrada weathering (assuming a period of normal polarity). The more probable explanation is, however, that this is a lithologic phenomenon similar to that described by Helsley and, in this case, a result of higher concentrations of red pigment since the intensity changes occur at lithologic breaks (base of Redonda Member and below the porous white sandstones). They do not, however, coincide with reversal boundaries (Figs. 2 and 3). This interpretation is supported by the low intensity of the middle normal compared with higher intensities of the upper normal and the normal and reversed intervals of the unnamed shale member. In the unnamed shale member these higher intensities are a result of a combination of more pigment and, in some cases, different magnetic mineralogy.

In addition to the anomalous zone in the upper normal interval the next nine samples above this zone have declinations within 30 degrees or less of due north but have inclinations which are negative. These inclinations progress with each step in demagnetization from initially positive inclinations to increasingly negative ones. Going stratigraphically upward, the nine inclinations pass (at 550° C) from $-1.6°$ to $-26.7°$ to $-0.3°$, then above the zone inclinations become positive again. Below the zone inclinations are also shallower than average but are not negative. Special care was taken to avoid demagnetizing stratigraphically consecutive groups of samples in the same demagnetization run and most of the demagnetized samples in the anomalous interval exhibit the anomalous directions at more than one demagnetization temperature. This indicates the negative inclinations are not the result of an extraneous field during cooling or of surement error. They must be considered a reflection of a directional excursion of the Upper Triassic field produced by a secondary or nondipolar term. This would also explain the low intensity in the anomalous zone as a fundamental property of the systematic deviation (Lawley, 1970) or as incompletely demagnetized samples from a reversed interval rather than as a lithologic phenomenon.

In establishing the reversal sequence proposed in this paper it should be pointed out that because of the high scatter within the upper reversed interval, and the sampling interval, small normal zones less than 1.5 m thick exist without being differentiated from the scatter. In fact, we considered the two short normal intervals within the upper reversed interval as scatter until we found one of them in the Redonda Mesa section. Close examination of the movement of the individual samples revealed that all of them had moved toward the Triassic normal direction during

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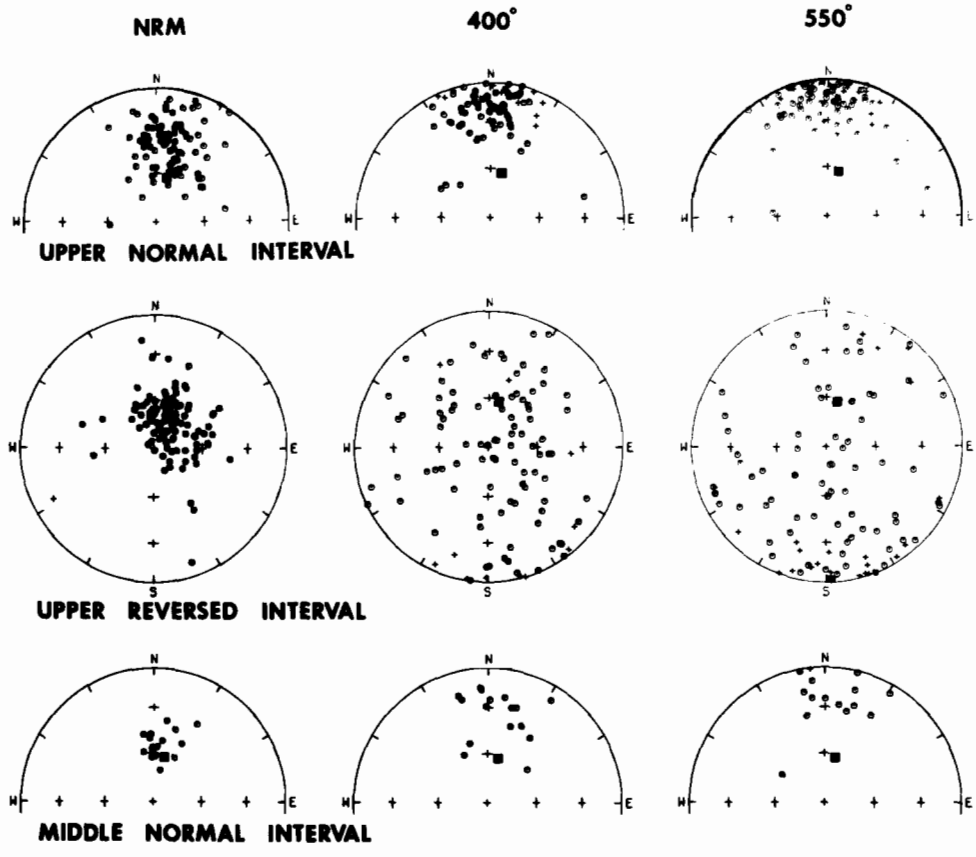


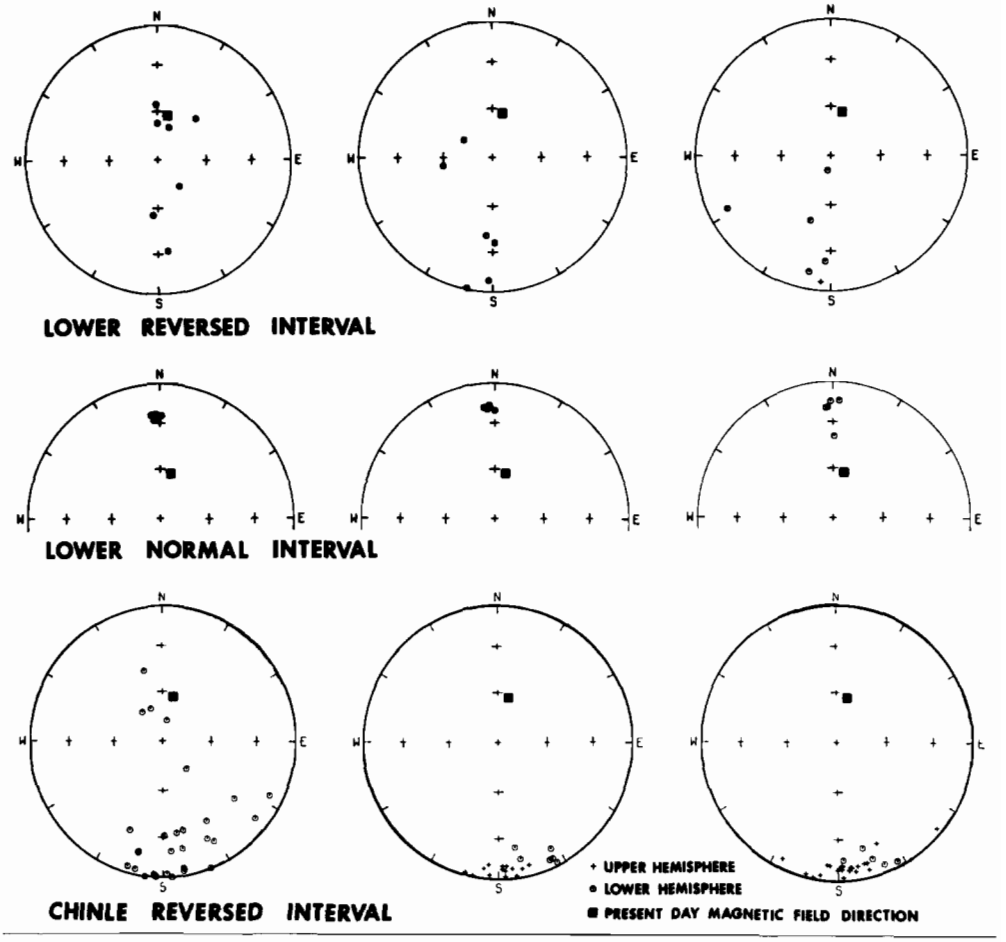
Figure 5. Schmidt net (equal area) plots of total group movement with temperature for each polarity interval.

each demagnetization step. Although several of the samples still exhibit inclinations close to 30°, this behavior was deemed to warrant their separation. Finally, in addition to the six polarity intervals discussed, additional work indicates that there is another normal and another reversed zone below the long reversed interval in the unnamed shale member. The lowermost normal interval appears to be 4.6 to 10.7 m thick. At present, the thickness is uncertain because of the difficulty of obtaining shale samples which are sufficiently well indurated to undergo thermal demagnetization.

POLE POSITIONS

Pole positions for each of the polarity intervals, average positions for normal and reversed groups, and two means for approximately the upper half of the Chinle Formation at Montoya, New Mexico, are given in Table 1. Average intensity and direction of magnetization, number of samples involved in the statistical treatment (N), resultant vector length of the summed sample determinations (R), precision parameter (k), half angle of the cone of confidence (α_{95}), circular standard deviation (δ), paleocolatitude of the sampling site, paleolongitude and paleolatitude of the paleomagnetic-pole, and semi-axes of elliptical error about the pole (dp, dm) have also been included. The pole positions for each of the intervals and the mean calculated from the various intervals for the Chinle are shown in Figure 6, along with the better demagnetized data for the Triassic of North America. In cases of duplicate work on the same formation or group, the study with the largest sampling has been used (other factors being equal). These pole positions for units other than the Chinle are listed in Table 2. All the data for the

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Upper Triassic have been combined with the well demagnetized data from the upper normal interval of the Redonda and the long reversed interval in the unnamed shale member of the Chinle to obtain an average paleomagnetic pole position for the Upper Triassic of North America at 89° E. long, 62.5° N. lat. This pole lies approximately 5.5° great-circle distance west of the pole calculated by Beck (1965) for the whole of the Triassic.[1]

The poles for the lower reversed and lower and middle normal intervals of this study should be considered statistically invalid because of the small number of samples involved and are given for internal comparison only. In calculating the Upper Triassic pole the upper reversed interval has not been used because of the many samples (36 of the 47 samples with incl >30) within it which appear to retain secondary magnetizations.

Average pole positions for the normal and reversed intervals (Table 1) are 8 degrees apart and are statistically indistinguishable with α_{95}'s of 8.9° and 18.3°, respectively.

The mean pole position for the Chinle Formation was calculated from 20 averages of 10 samples each; 40 samples were discarded as being incompletely demagnetized. It lies on the lower left edge of the triangle defined by the other Triassic demagnetized data, and is about seven degrees southwest of the mean Upper Triassic pole position defined by this study.

The Chinle mean paleomagnetic pole position is 5 to 12 degrees west of the Lower Triassic

[1] It is approximately the same distance southwest from the main Triassic pole calculated by Phillips and Forsyth (1972) at 66° N. lat, 98° E. long.

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TABLE 1. POLE POSITIONS AND STATISTICAL ANALYSIS FROM SELECTED THERMALLY (550° C) DEMAGNETIZED SAMPLES

Polarity interval	Average decrease	Average increase	Average intensity	N	R	k	α_{95}	δ	Colatitude	Pole longitude	Pole latitude	dp	dm
						Interval means							
Upper normal	357.8	9.7	.325E-04	71	67.82	22.01	3.67	17.15	85.1	80.42 E.	59.73 N.	1.88	3.71
Upper reversed	176.9	-1.7	.768E-05	67	52.00	4.40	9.39	38.34	90.9	98.48 W.	55.65 S.	4.70	9.39
Middle normal	5.8	15.9	.126E-04	12	11.46	20.53	9.81	17.12	81.9	63.55 E.	62.47 N.	5.20	10.10
Lower reversed	188.7	10.7	.679E-05	6	5.23	6.50	28.40	29.00	84.6	117.19 W.	48.67 S.	14.60	28.80
Lower normal	359.0	17.4	.125E-04	4	3.99	404.80	4.57	3.49	81.1	78.23 E.	63.81 N.	2.45	4.73
Chinle reversed	171.8	-4.7	.471E-04	31	29.60	21.56	5.69	17.17	92.3	89.05 W.	56.37 S.	2.86	5.71
						Grouped Means							
Normal intervals	0.8	14.4		3	2.99	193.1	8.90	4.76	82.7	74.24 E.	62.19 N.	4.66	9.11
Reversed intervals	179.1	1.4		3	2.96	46.52	18.30	9.70	89.3	102.41 W.	54.17 S.	9.15	18.30
All intervals	359.9	6.5		6	5.89	45.25	10.10	11.00	86.7	76.11 E.	58.17 N.	5.08	10.10
Ten sample	358.3	5.6		20	19.69	62.09	4.17	10.02	92.8	79.12 E.	57.66 N.	2.10	4.19

pole positions determined by Larochelle and Currie (1967), Robertson (1967), Helsley (1969), and McMahaon and Strangway (1968). Pole positions for the upper normal, upper reversed, and Chinle reversed intervals are each statistically indistinguishable from one or more of these determinations. The close similarity of the pole positions for these intervals and the Chinle mean to the Lower Triassic pole determinations indicates there was little, if any, north-south continental drift or polar wander relative to North America from the base of the Triassic through the period of recorded Chinle deposition in eastern New Mexico.

Helsley (1971) stated that John Graham has indicated that his early nondemagnetized data should no longer be used for pole determinations or comparisons. Because of the uncertain stability of the data, this opinion should be extended to much of the early work on the Chinle, although some of the data (for example, Collinson and Runcorn's 1960 determinations for the Chinle at Moab) may be fairly good. This paper does not consider any of the previous undemagnetized data. The pole position determined by Currie and Larochelle (1969) for the Mistatin volcanics has not been used, even though the data were demagnetized, because they presumed from the pole position that the age was wrong. Similarly, one could consider the age correct and regard the pole position as representing incomplete demagnetization (high stability secondary component) or a period of anomalous or unaveraged field. In either case, as the determination was suspect, it was not used in the calculation of a mean Upper Triassic pole. This exclusion may be unwarranted, as may the inclusion of the data from only two of the polarity invervals from the Chinle. One might argue for the inclusion of the upper reversed zone in the Redonda Member on the basis of its closeness to the mean. Unfortunately, it probably still contains some secondary component as evidenced by the remaining scatter at 550° C in Figure 5. Inclusion of the two is justified largely by their good grouping, higher intensity, and closeness to the mean direction, which changes by less than 1 degree on the exclusion of the two intervals.

De Boer (1968) attributed the northward streak of published Triassic poles to rapid polar wander or continental drift in latest Upper Triassic time. If this did occur, the failure of the Chinle data to exibit such a northward trend may indicate they were deposited and magnetized before the motion took place. Recent potassium-argon dating of igneous rocks in the Appalachians by Armstrong and Besancon (1970), and Foland and others (1971) allows more time for the northward shift to occur, and indicates the rate was probably less than the 25 cm per yr proposed by de Boer.

Much of the data cited by de Boer (1968) is

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TABLE 2. DEMAGNETIZED TRIASSIC POLE POSITIONS

Unit	Locality	Rock type	Pole Position	Reference
		Upper Triassic		
Diabase dike	Nova Scotia	Igneous	98 E., 69 N.	Larochelle and Wanless, 1966
Diabase dikes	SE. Pennsylvania	Igneous	105 E., 62 N.	Beck, 1965
Holyoke lava Lava in Granby tuff	Massachusetts	Igneous	88 E., 55 N.	Irving and Banks, 1961
Hampden Basalt	Connecticut	Igneous	92 E., 73 N.	deBoer, 1968
Mt. Carmel sill	Connecticut	Igneous	87 E., 61 N.	deBoer, 1968
West Rock dike	Connecticut	Igneous	89 E., 63 N.	deBoer, 1968
Talcott flows	Connecticut	Igneous	78 E., 51 N.	deBoer, 1968
North Mtn. flows	Nova Scotia	Igneous	77 E., 78 N.	deBoer, 1968
Cheshire dikes	Connecticut	Igneous	103 E., 76 N.	deBoer, 1968
Newark Group	New Jersey	Igneous and sedimentary	108 E., 63 N.	Opdyke, 1961
Holyoke lava	Connecticut	Igneous	83 E., 60 N.	deBoer, 1968
Deer Field flow	Massachusetts	Igneous	80 E., 44 N.	de Boer, 1968
Upper normal interval Redonda Member of Chinle Formation	New Mexico	Sediments	80 E., 60 N.	This study
Long reversed interval in unnamed shale member of the Chinle Formation	New Mexico	Sediments	91 E., 56 N.	This study
Mean Upper Triassic: 89 E., 62.5 N. $\alpha_{95} = 5°$				
		Lower Triassic		
Moenkopi Formation	Western Colorado	Sediments	98 E., 57 N.	Helsley, 1969
Manicouagan Group	Quebec	Igneous	88.5 E., 60.5 N.	Larochelle and Currie, 1967
Manicouagan Group	Quebec	Igneous	89 E., 57 N.	Robertson, 1967
Maroon Formation	Colorado	Sediments	101 E., 61 N. 99 E., 52 N.	McMahaon and Strangway, 1968

from lavas and, therefore, each determination samples only a small period of time, correspondingly allowing less chance of averaging out short term variations in the earth's magnetic field. On the basis of their radiometric work, Armstrong and Besancon (1970) proposed that the magnetic field of the earth was fluctuating rapidly during the late Upper Triassic. Although they may have been referring to a major polar shift caused by continental drift or polar wander, they imply, and the possibility certainly exists, that the northern determinations might represent a sampling of the short term variation of the magnetic field and are not due to continental drift or major polar wander. This possibility is reinforced by the findings of Steiner and Helsley (1972), whose pole determinations for the Triassic-Jurassic Kayenta Formation and Upper Jurassic Summerville Formation at 98° E. long, 59° N. lat–84° E. long, 59° N. lat (Kayenta), and 86° E. long, 70° N. lat (Summerville) are relatively close to the mean Upper Triassic pole position determined by this study. This indicates that the mean pole position for North America moved very little (if at all) throughout the Triassic if the pole determinations for the Kayenta and Summerville are truly representative. This would rule out the possibility of correlating the Chinle Formation with the Triassic formations of the eastern United States on the basis of pole position alone.

De Boer (1968) argued against the retention of secondary magnetization in the samples involved in the more northern pole determinations, citing high field AC demagnetization experiments and number of independent investigators involved. Some of these northern poles represent determinations based on undemagnetized samples; for example, Cox and Doell's (1960) position for the Shinarump (79° N. lat, 90° E. long) based on Graham's (1955) data. This cause for the streaking is also a possibility if the samples contain a high stability secondary component, and if either incomplete averaging or incomplete demagnetization are actually responsible for the northward streak, then the mean pole position for the Upper Triassic should move even closer to the data for the Lower Triassic. Similarly, inclusion of an average position

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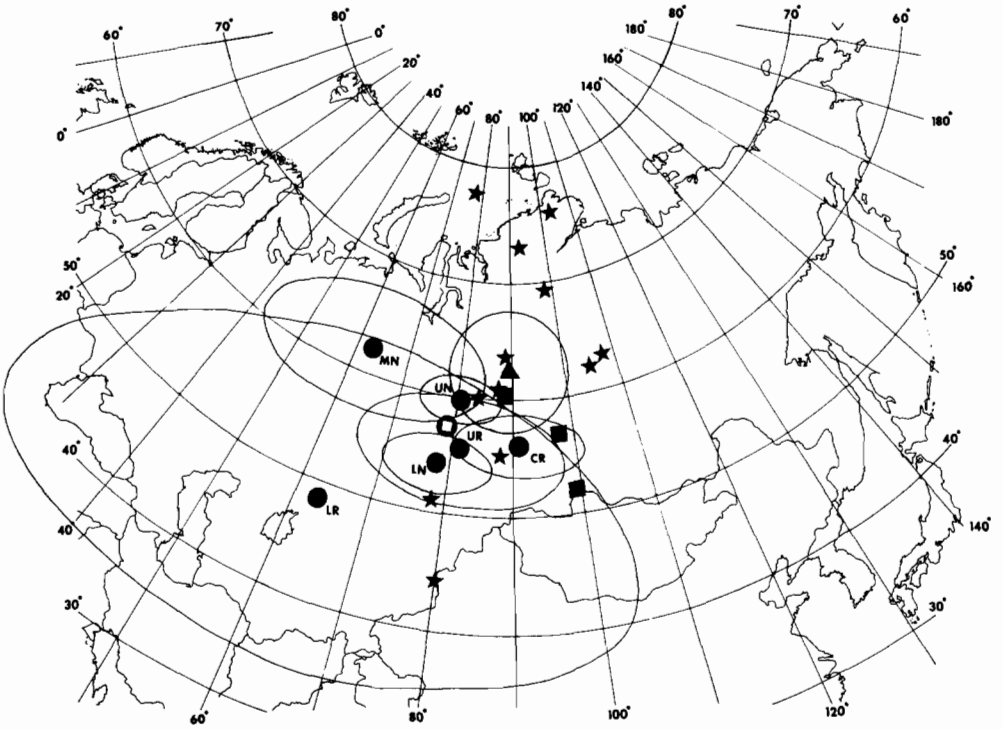


Figure 6. Map of Asia showing selected Triassic poles from previous publications compared with poles from polarity intervals described in this study. Solid circles = polarity intervals of this study with dpdm confidence ellipse. Solid circled square = mean Chinle pole (average of twenty 10-sample averages). Stars = previous Upper Triassic pole determinations based on demagnetized data (duplicated work not included). Squares = published demagnetized Lower Triassic determinations. Triangle = mean Upper Triassic pole position (with α_{95}) determined from upper normal and "Chinle Reversed" intervals of this study and the other Upper Triassic determinations shown in this figure. All pole positions are for North America only.

(92° E. long, 44° N. lat), which we calculated for North Carolina Upper Triassic sediments from values determined by Bowker (1960) and cited in de Boer (1967), would also move the mean Triassic pole toward the Lower Triassic determinations.

DISCUSSION OF CONFLICTING POLARITIES IN CHINLE AND NEWARK

The presence of a number of reversals in the upper half of the Chinle Formation in eastern New Mexico, and the almost complete absence of reversed polarity zones in the Upper Triassic of the eastern United States, are disconcerting when the close correlation of the two units on the basis of vertebrate paleontology (Colbert and Gregory, 1957; Schaeffer and Mangus, 1970) and paleobotany (Bock, 1969) is considered. Recently, McElhinny and Burek (1971) and Ash (1969) have suggested that some or all of the Chinle is older than the Newark Group in an attempt to explain paleomagnetic and paleobotanic differences between the two units.[2] The temporal placement of the Chinle relative to the Newark (McElhinny and Burek, 1971) should be regarded as tentative. This placement is limited by a number of paleontologic factors, and we believe that the complete mismatch proposed by McElhinny and Burek (1971), which places the Chinle in the Middle Triassic, is improbable.

[2] Ash (1972) reports that there are now three known species in common between the Chinle and the Newark of the northeastern U.S. This lessens the floral discrepancies that existed between the two units at the time of publication of his 1971 paper and supports the partial temporal overlap in depositional times proposed in our paper.

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Assigning the Chinle a Middle Triassic age essentially disregards most existing paleontologic data, particularly that of Colbert and Gregory (1957), Stovall and Savage (1939; who suggest the placement of the Redonda high in the North American Triassic sequence), Baird and Patterson (1967), Hope and Patterson (1969), and Gottesfeld and Kremp (1968). The work of Baird and Patterson (1967) and Hope and Patterson (1969) suggests the base of the Triassic section in North Carolina is correlatable with lower-middle portions of the Chinle in Arizona. The findings of Gottesfeld and Kremp (1968) suggest a Carnian age at most for the Petrified Forest Member of the Chinle Formation in Petrified Forest National Monument, Arizona. This is the oldest possible age and the sediments studied by Gottesfeld are probably much younger (Gottesfeld, 1971, oral commun.)

McElhinny and Burek's (1971) proposed long period of Upper Triassic normal polarity rests essentially on three sedimentary sequences: the Newark, the Red Sandstone of the Zambezi Valley (Opdyke, 1964), and Italian sediments described by Zijderveld and deJong (1969). In all cases the temporal relation of these sediments to the Chinle is uncertain. The Red Sandstone Formation of the Zambezi Valley is unfossiliferous (Opdyke, 1964), and sampling of the Newark Group sediments is hampered by long non-red bed intervals. Zijderveld and deJong (1969) believed their findings from the Italian sediments were a result of the present magnetic field and should be discarded. Given the general contemporaneity of radiometric age dates cited by Brock (1968), Armstrong and Besancon (1970), Larochelle and Wanless (1966), Currie and Larochelle (1969), Erickson and Kulp (1961), and Carmichael and Palmer (1968) for Upper Triassic rocks of both normal and reversed polarities, an alternate hypothesis more in line with paleontological evidence would suggest a partial temporal overlap of the Chinle and Newark. This, of course, makes necessary an assumption that portions of some sections of the Lower Newark were deposited during periods of reversed polarity, and requires a more complex interpretation than a simple normal-reversed-normal polarity history for the African igneous rocks. The presence of the reversed zones reported from the Lower Newark in Bowker's (1960) dissertation supports this hypothesis.

Paleontologic and paleomagnetic data are still insufficient to determine with certainty the existence or amount of overlap in the periods of deposition of the Upper Chinle and Lower Newark. Paleontologic and spatial factors tend to rule out most other situations that would explain the difference in the observed polarity histories of the Chinle Formation in the west and the Newark Group in the northeast. Galton (1971) has pointed out the apparent contemporaneity of the Navajo Sandstone and the Portland Arkose, and Walker (1968) suggested the near contemporaneity of the Dinosaur Canyon Sandstone Member of the Moenave with the Long Meadow Sandstone Member of the Portland Arkose. These correlations, and the study by Steiner and Helsley (1972) dealing with the Kayenta, suggest the Chinle in the Colorado Plateau is not younger than the Newark Group but is somewhat older. The Redonda Member of the Chinle Formation is almost certainly older than the upper two formations of the Glen Canyon Group. This means one cannot explain the observed polarity differences between the Chinle in eastern New Mexico and the Newark Group by saying the section sampled in this study is younger than the Newark. Likewise, paleontology rules out extremely rapid deposition of either the Chinle Formation or the Newark Group.

Early hematite cementation and acquisition of remanence in the Redonda are supported by the observation that the remanence was being acquired at a rate sufficient to record the periods of reversed polarity. Additionally, Creer (1971) has published a Mesozoic reversal chronology established by Pecherski (1970) that shows a period of mixed polarity (Finishnaja zone) from the upper Norian to the late Rhaetian. This agrees very well with the present ages given for the Redonda Member and underlying unnamed shale member, and supports early acquisition of remanence in the upper portion of the Chinle Formation in eastern New Mexico. Creer states that McElhinny and Burek's Nuanetsi zone corresponds to Pecherski's Maseru and Levo-Kedonskaja zones. Correlation of the Stormberg red beds (which overlie the Nuanetsi lavas) with the Kayenta (Galton, 1971), along with age assignments for the Newark Group (Reeside and others, 1957), suggest that correlation of the Nuanetsi with the Finishnaja is more appropriate. This leads to a second possibility for the observed discrepancy. It is probable

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that much of the upper portion of the Newark Group in the northeastern United States was deposited in the long normal interval between the Finishnaja interval of mixed polarity and the Levo-Kedonskaja reversed interval. The discrepancy between the Newark Group and the upper portion of the Chinle in eastern New Mexico could be easily explained if only the lower sections of the Newark Group were contemporaneous with the Chinle and those contemporaneous sections had acquired their remanence 4 to 5 m.y. or so after deposition. This is entirely possible, considering that Walker (1967) found a 1 m.y. time lag between deposition and reddening in Pliocene sediments from Baja California. In fact, de Boer (1972, Oral commun.) has documented such a time lag in his studies of Newark Group sediments and agrees with this correlation between the Newark and Chinle.

SUMMARY

The Chinle Formation of eastern New Mexico has at least eight and possibly eleven polarity intervals in its upper half alone. In addition, there is some evidence for a possible short period of anomalous field behavior during the deposition of the upper normal interval. The existence of these intervals raises a question over the validity of present paleontologic correlations between the Newark Group of the eastern United States and the Chinle. Attempts by McElhinny and Burek (1971) to explain the absence of reversed polarities in the Newark Group should be regarded with skepticism.

Armstrong and Besancon (1970) concluded from their potassium-argon studies that volcanics from the Newark Group are unsuitable for defining a point in the geologic time scale. It is suggested that until further paleomagnetic and paleontologic work have been done in the western United States to clarify the detailed temporal relations between the Chinle Formation and the Newark Group, the Newark Group should not be used to define a portion of a reversal sequence chronology, such as McElhinny and Burek's, because of the possibility that some of the Newark Group's sediments were deposited during periods of reversed polarity.

This paper represents the first work done in the Chinle Formation since demagnetization of samples has become common. It is hoped that it will encourage further detailed paleomagnetic investigations in the Chinle Formation of the western United States. Detailed paleomagnetic-stratigraphic studies in the lower portions of the Chinle and in the overlying Wingate Formation would go far in answering the seeming ambivalence between present paleomagnetic and paleontologic data, particularly if they were combined with extensive palynologic or other paleontologic studies.

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APPENDIX. REDONDA MESA SECTION

The stratigraphic section collected at Redonda Mesa serves to check the reversal sequence derived at Lucianna Mesa 35 km to the west and provides information about the deposition of the Redonda Member of the Chinle Formation. In general, sampling, demagnetization, and measurement techniques and instrumentation were the same as those used in examining the Lucianna Mesa Section. No Curie balance or thin section studies were made on the Redonda Mesa samples and demagnetization was carried out initially only at 500° C; 49 samples were collected from this locality.

Stratigraphic plots of natural remanent magnetization and 550° C thermal demagnetization measurements are shown in Figure 7. The base of the sampled section coincides with the base of the Redonda Member of the Chinle Formation. The section terminates just below the base of a massive (thicker than 10 m) non-red sandstone within the Redonda Member. Magnetically, the section begins just below what we have termed the lower reversed interval and continues midway up into the upper reversed interval.

Demagnetization at 600° C of the 12 lowest samples in the section was undertaken in an attempt to more accurately define the normal and reversed intervals. The results of this step were inconclusive, with some samples which had previously

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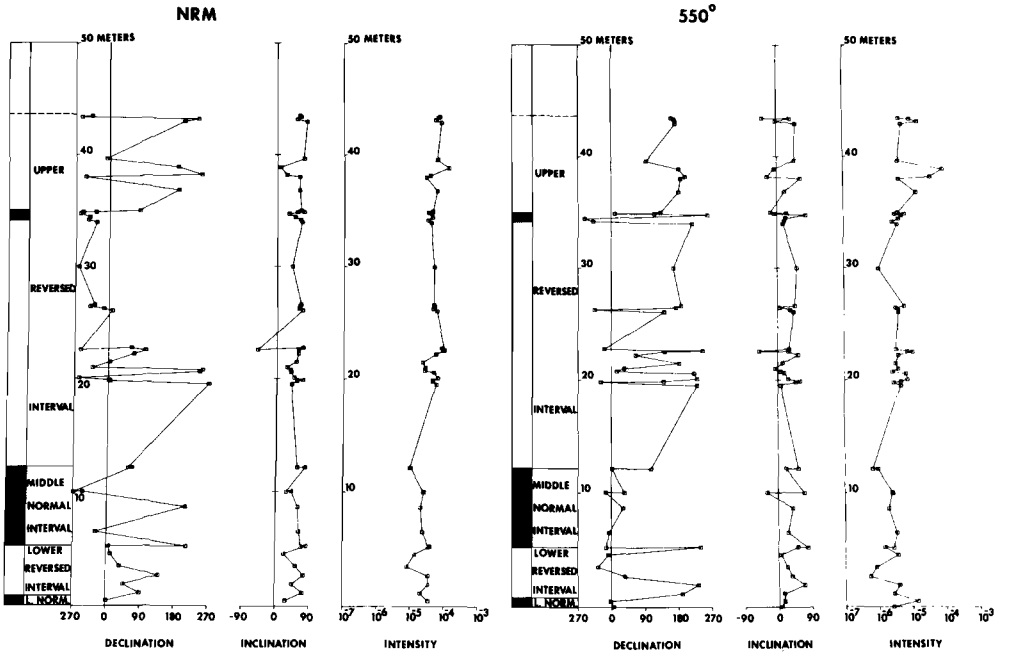


Figure 7. Stratigraphic plots of NRM and 550° C demagnetized sample declination, inclination, and intensity for Redonda Mesa section. Base of section coincides with base of the Redonda Member.

moved in a reversed direction returning to alignments in the direction of the normal and present-day field directions. Most of the samples probably retain some secondary magnetization after the 550° C demagnetization step. Some of this secondary component is probably due to the present field. In addition, three samples moved in a direction similar to samples TD 189 and TD 161 in Figure 4A during the 550° C demagnetization step then turned toward the northwest at a very shallow inclination at 600° C. These samples center in and around the upper portion of the lower reversed interval as it is shown in Figure 7 and may represent samples deposited during a period of reversed field and then influenced by the succeeding period of normal polarity. This would explain their deviation and secondary(?) Triassic normal direction. A second and possibly more likely alternative is that the lower reversed interval is just very thin and is actually represented by only the lower two samples in the interval as it is now chosen in Figure 7. This is a strong possibility, as one of the samples in the interval and the two below show characteristic normal movement with no hint of reversed behavior. Sampling at a closer spacing will be necessary to adequately distinguish between these possibilities.

Besides the four main polarity intervals two other features occur in both the Lucianna and Redonda Mesa sections. The short normal interval within the upper reversed interval probably corresponds to the upper of the two short questionable normals in the same interval on Lucianna Mesa. Likewise, the intensity jump at 39 m corresponds to a similar jump at 96 m on Lucianna Mesa. At present we do not know if this intensity jump is due to lithologic or paleofield intensity changes. The general hand-specimen appearances of the beds involved are no different than their lower intensity counterparts but, because the rocks are very fine grained, subtle lithologic distinctions should not be ruled out without extensive examination.

In conclusion, comparison of the sections at Lucianna and Redonda Mesas confirms the existence of a portion of the reversal sequence derived at Lucianna Mesa. It also indicates that deposition of the Redonda lithology began somewhat earlier at Redonda Mesa than at Lucianna Mesa to the west. Furthermore, it shows that, within the accuracy defined by the sampling interval and the magnetic stability of the rocks investigated, depositional rates of the lower portion of the Redonda Member of the Chile Formation at Lucianna and Redonda Mesas are comparable.

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